Chapter 5: Differentiation

Section: L'Hospital's Rule

Based on Walter Rudin's *Principles of Mathematical Analysis*

Outline

Motivation

The Problem: Indeterminate Forms

- Many limits have the form $\lim_{x \to a} \frac{f(x)}{g(x)}$ where $f(x) \to 0$ and $g(x) \to 0$

$$\frac{0}{0} \quad \longleftarrow$$ indeterminate: the value could be anything

The Problem: Indeterminate Forms

- Many limits have the form $\lim_{x \to a} \dfrac{f(x)}{g(x)}$ where $f(x) \to 0$ and $g(x) \to 0$
- The same trouble appears when $g(x) \to +\infty$: the form $\dfrac{\text{something}}{\infty}$

$$\frac{0}{0} \qquad\qquad \frac{*}{\infty}$$

The Idea: Compare Rates of Change

- Near a, both f and g are governed by their derivatives
- Heuristic linearization (using $f(a) = g(a) = 0$): $f(x) \approx f'(a)(x-a)$, $g(x) \approx g'(a)(x-a)$

$$\frac{f(x)}{g(x)} \approx \frac{f'(a)(x-a)}{g'(a)(x-a)} = \frac{f'(a)}{g'(a)}$$

A Classic Example: $\dfrac{\sin x}{x}$

Example (trig derivatives previewed from Chap. 8)

$$\lim_{x\to 0}\frac{\sin x}{x} = \lim_{x\to 0}\frac{\cos x}{1} = 1$$

The Rule

Theorem 5.13: L'Hospital's Rule

Theorem 5.13

Suppose f, g are real and differentiable in (a, b), with $g'(x) \neq 0$ there and $-\infty \leq a < b \leq +\infty$. If

$$\frac{f'(x)}{g'(x)} \to A \quad (x \to a) \tag{13}$$

and either

$$f(x) \to 0, \quad g(x) \to 0 \quad (14) \qquad \text{or} \qquad g(x) \to +\infty \quad (15)$$

as $x \to a$, then

$$\frac{f(x)}{g(x)} \to A \quad (x \to a). \tag{16}$$

Reading the Two Cases

Case (14): the $\frac{0}{0}$ form

$f \to 0$ and $g \to 0$.

$$\frac{f}{g} \to A$$

Case (15): the $\frac{*}{\infty}$ form

$g \to +\infty$ (the top is *unrestricted*).

$$\frac{f}{g} \to A$$

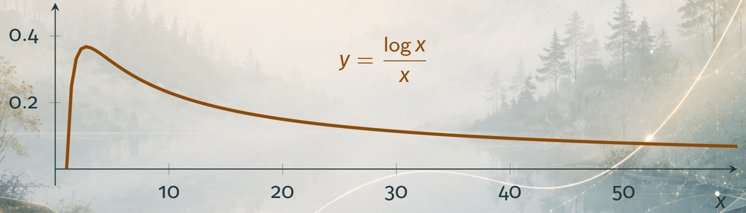
- In case (15) we need nothing about f — only that $g \to +\infty$
- Both cases share the same conclusion and the same hypothesis (13)

Case (15) in Action: $\dfrac{\log x}{x}$

Example (log derivative previewed from Chap. 8)

As $x \to +\infty$, both $\log x \to +\infty$ and $x \to +\infty$:

$$\lim_{x \to +\infty} \frac{\log x}{x} = \lim_{x \to +\infty} \frac{1/x}{1} = 0.$$



Remarks on the Statement

Variants and conventions

- Analogous statements hold as $x \to b$, or if $g(x) \to -\infty$ in (15)
- Limits are taken in the extended sense of Definition 4.33, so A may be $\pm\infty$

Why $g' \neq 0$ matters: a derivative has the intermediate value property (Theorem 5.12), so g' cannot be positive at one point and negative at another without vanishing in between. Hence $g' > 0$ on all of (a, b), or $g' < 0$ on all of (a, b) — so g is strictly monotonic and $g(x) - g(y) \neq 0$, keeping every denominator in the proof safe.

Proof

Proof Strategy: Theorem 5.13

Goal: Show $f(x)/g(x) \to A$ as $x \to a$.

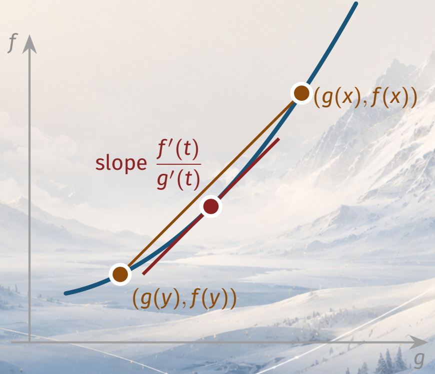
Approach (squeeze from both sides):

1. Pick any $q > A$ and an intermediate r with $A < r < q$; hypothesis (13) confines f'/g' below r near a
2. The generalized MVT (Theorem 5.9) transfers this bound to a ratio of *function values*
3. Feed in case (14) or case (15) to conclude $f(x)/g(x) < q$ near a
4. Symmetrically, any $p < A$ gives $p < f(x)/g(x)$ near a
5. Squeeze: $p < f/g < q$ for all $p < A < q \Rightarrow f/g \to A$

The Engine: Generalized Mean Value Theorem

Recall: Theorem 5.9

For $a < x < y < c$, there is $t \in (x, y)$ with $\dfrac{f(x) - f(y)}{g(x) - g(y)} = \dfrac{f'(t)}{g'(t)}$.



Proof of Theorem 5.13

Step 1: Confine the derivative ratio

Assume $-\infty \leq A < +\infty$. Choose q with $A < q$, then r with $A < r < q$. By (13) there is $c \in (a, b)$ such that

$$a < x < c \implies \frac{f'(x)}{g'(x)} < r. \tag{17}$$

Proof of Theorem 5.13

Step 2: Apply the generalized MVT

If $a < x < y < c$, Theorem 5.9 gives $t \in (x, y)$ with

$$\frac{f(x) - f(y)}{g(x) - g(y)} = \frac{f'(t)}{g'(t)} < r. \tag{18}$$

Key step: the bound (17) on f'/g' now controls a ratio of function values.

Proof of Theorem 5.13

Step 3: Case (14), the $\frac{0}{0}$ form

Fix y and let $x \to a$ in (18). Since $f(x) \to 0$, $g(x) \to 0$:

$$\frac{f(y)}{g(y)} \leq r < q \qquad (a < y < c). \tag{19}$$

Proof of Theorem 5.13

Step 4: Case (15), the $g \to +\infty$ form

Fix y; choose $c_1 \in (a, y)$ so that $g(x) > g(y)$ and $g(x) > 0$ for $a < x < c_1$.

Multiply (18) by $\dfrac{g(x) - g(y)}{g(x)} > 0$:

$$\frac{f(x)}{g(x)} < r - r\,\frac{g(y)}{g(x)} + \frac{f(y)}{g(x)} \qquad (a < x < c_1). \tag{20}$$

Proof of Theorem 5.13

Step 4 (continued): let $x \to a$

In (20), y is fixed and $g(x) \to +\infty$, so the correction terms vanish. Hence there is $c_2 \in (a, c_1)$ with

$$\frac{f(x)}{g(x)} < q \qquad (a < x < c_2). \tag{21}$$

Both cases meet here: (19) and (21) give the same bound $\frac{f(x)}{g(x)} < q$ near a, for every $q > A$.

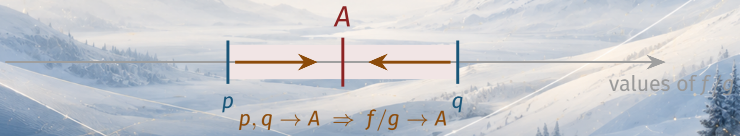
Proof of Theorem 5.13

Step 5: Symmetric lower bound and conclusion

If $-\infty < A \leq +\infty$, pick $p < A$; the mirror argument yields c_3 with

$$p < \frac{f(x)}{g(x)} \qquad (a < x < c_3). \tag{22}$$

So for all $p < A < q$, eventually $p < \dfrac{f(x)}{g(x)} < q$, hence $\boxed{\dfrac{f(x)}{g(x)} \to A}$ ■



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Key takeaways from this section:

- **Theorem 5.13 (L'Hospital):** if $f'/g' \to A$ and either $f, g \to 0$ or $g \to +\infty$, then $f/g \to A$

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- **Generality:** endpoints and A may be $\pm\infty$; the $g \to +\infty$ case needs no hypothesis on f

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- **Theorem 5.13 (L'Hospital):** if $f'/g' \to A$ and either $f, g \to 0$ or $g \to +\infty$, then $f/g \to A$
- **Engine of the proof:** the generalized mean value theorem (5.9) turns a derivative bound into a function-value bound
- **Generality:** endpoints and A may be $\pm\infty$; the $g \to +\infty$ case needs no hypothesis on f
- **Warning:** the rule fails for complex / vector valued functions (Example 5.18)

Coming up next: Derivatives of Higher Order and Taylor's Theorem.